

1. Consider:

```
x = torch.tensor([2.0], requires_grad=True)
y = x**2 + 3*x + 1
y.backward()
```

What is the value inside `x.grad`?

Explain how PyTorch computed it step-by-step.

2. What happens in this case?

```
x = torch.tensor([2.0])
y = x**2
```

```
y.backward()
```

Why does it fail?